Answers to Lecture Notes

MATH0029: Graph Theory and Combinatorics

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Apologia

This document contains worked answers and notes corresponding to the lecture notes of MATH0029: Graph Theory and Combinatorics, as taught by Dr. António Girão in 2025, written by a student taking the class. A lot of the answers given will focus more on getting the 'spirit' of the question correct, rather than stressing technical details of the proofs.

I've tried my best to check them all, but please note that these answers are **not** officially endorsed, so expect about $\lceil n/2 \rceil$ to be actually correct. Please email me with any mistakes you find.

If you would like to contribute by helping fill in blank questions, please let me know!

London, England Archie

Chapter 1

Basic Counting: Sets and Tuples

Question 1.1. How large is $\binom{[6]}{3}$?

Answer: From Lemma 1.2(iv) we have that $|\binom{[6]}{3}| = \binom{6}{3} = 20$

Question 1.2. For $A \subseteq \mathbb{N}$ define $S(A) = \Sigma_{a \in A} a$. For how many sets $A \in \binom{[6]}{3}$ is $S(A)$ odd?

Answer: $\binom{[6]}{3} = \{A \subseteq [6] : |A| = 3\}$. In order for $S(A)$ to be odd, we need an text odd number of odd elements (since odd + odd + even = even), and at least one odd number (as even + even + even = even). This then gives 2 possible configurations of sets:

$$\{odd, odd, odd\} \text{ or } \{odd, even, even\}$$

There are $\binom{3}{3} = 1$ of the first type, and $\binom{3}{1}\binom{3}{2} = 3 \cdot 3 = 9$ of the second type, thus giving 10 total sets where $S(A)$ is odd.

Question 1.3. Let $n \in \mathbb{N}$. How large are the following family of sets:

$$\mathcal{F}_1 = \{A \subseteq [n] : 1 \in A\}$$
$$\mathcal{F}_3 = \{A \subseteq [n] : |A \cap [3]| \geq 2\}$$

Answer: For the first part, a quick observation will lead to conjecturing $|\mathcal{F}_1| = 2^{n-1}$ To prove this, proceed by induction on $n \in \mathbb{N}$. The base cases is obviously true for $n = 1$, so assume true for $n = k$ and consider $\mathcal{F}_1$ when $n = k + 1$. Observe

$$\begin{aligned}\mathcal{F}_1 &= \{A \subseteq [n] : 1 \in A\} \\ &= \{A \subseteq [k] \cup \{k+1\} : 1 \in A\}\end{aligned}$$

From the inductive hypothesis, we already know that there are 2^{k-1} sets containing 1. By appending a the number $k + 1$ on the end of all of these sets, we get 2^{k-1} additional distinct sets containg 1, and no more since order is irrelevant in sets. Hence $|\mathcal{F}_1| = 2^{k-1} + 2^{k-1} = 2 * 2^{k-1} = 2^{(k+1)-1}$.

For the second part, we can count how many elements of [3] we get in A. Either we get exactly two, or all 3. If you fix which elements are in A, given we take two or three, the remaining $n-3$ elements of $\{4, \ldots, m\}$ are free, giving 2^{n-3} choices each time. This gives us:

$$|\mathcal{F}_{\ni}| = (\binom{3}{2} + \binom{3}{3})2^{n-3} = 4 \cdot 2^{n-3} = 2^{n-1}$$

Question 1.4. How many sets in $\binom{[6]}{3}$ do not contain two consecutive integers?

Answer: There's only 20 sets, so its actually quite easy to just list them. Starting with 1, as the first entry, there must be a gap of at least two between each element of the set, giving us $\{1, 3, 5, \}, \{1, 3, 6, \}, \{1, 4, 6\}$. Now starting with 2, you can only get $\{2, 4, 6\}$. You cannot start with 3 since your last element would have to be at least 7, which is not in the set. Hence there are 4 such sets.

Question 1.5. Let $n \in \mathbb{N}$ be odd. Find a simple expression for the sum

$$\sum_{k>n/2} \binom{n}{k}$$

Answer: We have from the binomial theorem that $(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$ hence the sum of binomial coefficients is 2^n. Binomial coefficients are symmetric and there is no central term since n is odd (see: Pascal's Triangle), hence:

$$\sum_{k>n/2} \binom{n}{k} = 1/2 \sum_{k=0}^{n} \binom{n}{k} = 2^{n-1}$$

Question 1.6. Let $n \in \mathbb{N}$ for which $k \in \mathbb{N}$ is $\binom{n}{k}$ maximal? Is k unique?

Answer: We have (with a bit of work) that $\binom{n}{k} - \binom{n}{k-1} = \frac{n+1-2k}{n+1}\binom{n+1}{k}$. So for fixed n, we have that the sequence of binomials is increasing when $2k \leq n+1$ and decreasing when $2k \geq n+1$. Hence its maximum is attained when $k = \frac{n+1}{2}$. Since n is a natural number, this is only unique when n is odd. When n is even you can either take the floor or the ceiling.

Question 1.7. Give a combinatorial proof that if $0 \leq a, k \leq n$ are integers then

$$\sum_{b=0}^{k} \binom{a}{b}\binom{n-a}{n-b} = \binom{n}{k}$$

Question 1.8. Give a combinatorial proof that if $0 \leq k \leq n$ are integers then

$$\binom{n}{k} = \sum_{a=k}^{n} \binom{a-1}{k-1}$$

Question 1.9. Let $m, n \leq 1$ be integers. Consider the paths starting from $(0,0)$ in $\mathbb{Z}^2$ in which each step is from (a,b) to $(a+1,b)$ or from (a,b) to $(a,b+1)$.

1. How many such paths are there from $(0,0)$ to (m,n)
2. How many such paths are there from $(0,0)$ to (n,n) that always remain on or below the line $y = x$?

Question 1.10. Use induction to prove that for $n \geq 1$

$$\binom{2n}{n}\frac{1}{4^n} = \frac{1}{2n}\prod_{i=1}^{n-1}(1+\frac{1}{2j})$$

Show that for $n \in \mathbb{N}$

$$\frac{4^n}{2\sqrt{n}} < \binom{2n}{n} < \frac{4^n}{\sqrt{2n+1}}$$

Chapter 2

Graphs

2.1 Basic Definitions

Question 2.1. Prove that any graph of order $n \geq 2$ has two vertices of the same degree.

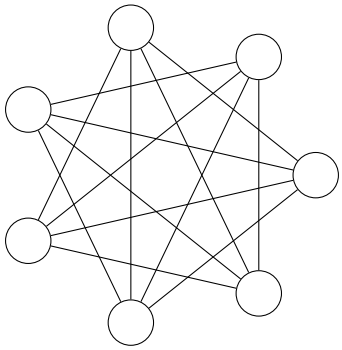
Answer: The maximum degree a vertex can have in any simple graph is $n-1$. Observe the degree sequence:

$$0 \leq d(x_1) \leq d(x_2) \leq \cdots \leq d(x_n) \leq n-1$$

Since there are n vertices, and only $n-1$ values the degree that any vertex can take, by the pigeonhole principle, two vertices must share the same degree.

Question 2.2. Can you (a) construct a 4-regular graph of order 7? If not, why not? (b) How about a 3-regular graph of order 7?

Answer: For (a), observe:



For (b), this is not possible as it violates the handshake lemma: the sum of degrees must be even, whereas $7 * 3 = 21$ is odd.

Question 2.3. Let G be a graph of order $n \geq 2$ in which every vertex has degree at least $(n-1)/2$. Prove that for every pair of distinct non-adjacent vertices $u, v \in V(G)$ there is a third vertex w such that $uw, vw \in E(G)$

Answer: Consider the neighborhoods of both vertices u and v. By assumption,

$$|\Gamma(u)| \geq \frac{n-1}{2}, \quad |\Gamma(v)| \geq \frac{n-1}{2}$$

and since u and v are non adjacent and we are working with simple graphs, we have that

$$\Gamma(u), \Gamma v \subseteq V(G) \setminus \{u, v\}$$

which has $n-2$ vertices. $|\Gamma(u)| + |\Gamma(v)| \geq \frac{n-1}{2} + \frac{n-1}{2} = n-1$. So by pigeonhole, $\Gamma(u)$ and $\Gamma(v)$ must share a vertex. Label that vertex w and the statement is proved.

2.2 Examples of Graphs

Question 2.4. Let $n \in \mathbb{N}$. Which of the graphs C_n, K_n, $K_{n,2n}$, $K_{n,n}$ and Q_n are regular? For those that are regular, calculate the degree of a vertex in the graph.

Answer: C_n: C_n is 2-regular.
K_n: K_n is n-1 regular.
$K_{n,2n}$: $K_{n,2n}$ is not regular as all the vertices in one partition have degree n and all the vertices in the other partition have degree $2n$.
$K_{n,n}$: $K_{n,n}$ is n-regular as each vertex from one partition is mapped to n vertices in the other partition.
Q_n: Q_n is n regular, as each vertex has n coordinates from which to differ from.

Question 2.5. Let $n \in \mathbb{N}$. What are the sizes of the graphs: C_n, K_n, $K_{n,2n}, K_{n,n}$, Q_n?

Answer: All of these can be worked out from the handshake lemma: divide the sum of vertex degree by 2.
$|C_n| = n$ as there are n vertices of degree 2.
$|K_n| = \frac{n(n-1)}{2}$ as there are n vertices of degree $n-1$
$|K_{n,2n}| = 2n^2$ as there are n vertices of degree $2n$, and $2n$ vertices of degree n.
$|K_{n,n}| = n^2$ as there are n vertices of degree n and then another n vertices of degree n.
$|Q_n| = \frac{2^n n}{2}$ as there are 2^n vertices of degree n.

Question 2.6. Let $1 \leq a \leq b$ be integers. If $K_{a,b}$ has order n what is the smallest and largest that the size of $K_{a,b}$ could be?

Answer: Obviously, the smallest is when $a = b = 1$. For the largest, you can solve via calculus, but intuitively, since $a + b = n$ and from before $|K_{a,b}| = ab$, everytime we increase the value of a, b must get smaller (and visa versa). Hence the product is maximised when a and b are balanced: $a = b = n/2$. So the largest size is $n^2/4$.

2.3 Subgraphs and isomorphisms

Question 2.7. For which $a, b, t \in \mathbb{N}$ does $K_{a,b}$ contain a copy of C_t?

Answer: I opt for a less formal answer in this case. Assume $a, b, t > 1$ since for $a = b = 1$ you can only get C_1 and C_1 is present for all $a, b > 0$. Consider the parity of t. When t is odd, we would need that two vertices of partition are adjacent, so t must be even. A cycle needs to alternate between the two sides, which means it must use the same number of vertices from each side. If a cycle is of length t, it therefore uses $t/2$ vertices from A and $t/2$ vertices from B, so we need enough vertices on each side to supply $t/2$ distinct ones. Hence we get $t/2 \leq min\{a, b\} \implies t \leq 2min\{a, b\}$.

Question 2.8. How many copies of $K_{1,2}$ are there in $K_{a,b}$ (where $a, b \geq 2$)?

Answer: For every vertex in partition A, we just pick 2 vertices in partition B, hence $a \cdot \binom{b}{2}$

Question 2.9. When does $K_{a,b}$ contain a cycle of length $a + b$?

Answer: From the earlier question, we need $a + b \leq 2min\{a, b\}$. WLOG $a \leq b$, then $a + b \leq 2a \implies b \leq a$. So only when $a = b$.

Question 2.10. How many copies of C_4 does Q_n contain for $n \geq 2$?

Answer: This was a coursework question!

2.4 Walks, paths, and connectedness

Question 2.11. What is the length of a longest path in each of the following graphs: $P_n, C_n, K_n, K_{a,b},$ and Q_n

Answer: P_n: the longest path is of length $n - 1$, obviously...
C_n: the longest path is of length $n - 1$, since any longer will revisit the starting vertex.
K_n: the longest path is again of length $n - 1$ as we visit all n vertices in any order.
$K_{a,b}$: From the previous question, the longest cycle is of length $2 * min\{a, b\}$, hence the longest path will be $2min\{a, b\} - 1$,
Q_n: Bit of a cheat but the graph is hamiltonian so the longest path will be $2^n - 1$.

Question 2.12. For a connected graph G define the path distance function

$$d : V(G)^2 \to \mathbb{Z}^+ \text{ by } d(u, v) = \text{the length of the shortest } u - v\text{-path}$$

Compute d explicitly for the graphs P_n, C_n, K_n, $K_{a,b}$, and Q_n

Answer: P_n: for $i, j \in P_n, d(i, j) = |i - j|$, since you must traverse all vertices between.
C_n:
K_n: for $i, j \in K_n, d(i, j) = 1$ if $i \neq j$, 0 otherwise.
$K_{a,b}$: for $i, j \in K_{a,b}, d(i, j) = 1$ if the vertices are in different partitions, and 2 if they are in the same partition.
Q_n: for $i, j \in Q_n, d(i, j) =$ the number of coords for which the two vertices differ

Question 2.13. Suppose G has girth at least 7 and minimum degree at least 50, prove that G must have order at least 122551.

Answer: Consider a vertex $v \in V(G)$. By assumption, this vertex has at least 50 neighbours. Consider an arbitrary neighbour w, by assumption, w must have at least 50 vertices. This is true for all neighbours of v, since if two w's shared a vertex we would have a cycle of length 4. This argument then applies again to 50 vertices adjacent to w, implying that we have at least $50^3 = 125000$ vertices.

2.5 Trees

Question 2.14. Show that sub-trees of a tree satisfy Helly-type property.

Answer: Coursework question!

Question 2.15. Show that if T contains a vertex of degree d, then it must contain at least d leaves. Is this tight?

Answer: Let T be a tree and let $v \in V$ have degree d, let the neighbours of v be $u_1, \ldots, u_d$. Removing v and its edges splits the graph into d connected components $C_1, \ldots, C_d$, each being a tree. Every finite tree has at least one leaf, choose a leaf $w_i \in C_i$. If C_i is the single vertex u_i, then $u_i = w_i$ is a leaf of T. If C_i has more than one vertex, then its leaf w_i is not u_i, so w_i has degree 1 inside C_i and is not adjacent to v. Therefore w_i has degree 1 in T, and thus is a leaf in T. $w_1, \ldots, w_d$ lie in different components C_i. So T has at least d leaves. Tight for $K_{1,d}$.

2.6 Euler Circuits

Question 2.16. Let G be a connected graph in which all vertices have even degree except two which have odd degree. Show that there is a walk in G using every edge exactly once.

Answer: Let the two vertices of odd degree be u, v. Create a new graph G' by adding a new edge between u and v (if they're adjacent, just add a parallel edge). Then G' is a connected graph wherein all vertices have even degree, and hence has an eulerian circuit C. If we remove the edge uv from C, what remains is an eulerian trail whose endpoints are u and v.

2.7 Bipartite Graphs

Question 2.17. Let $G = (V, E)$ be a bipartite graph with non-empty vertex classes X and Y. Show that if every vertex in X has degree $x \geq 1$ and every vertex in Y has degree $y \geq 1$, then $x|X| = y|Y|$.

Answer: This follows from a short double counting argument. Every edge of G has one endpoint in X and one endpoint in Y, so we have:

$$\sum_{v \in X} deg(v) = |E| = \sum_{v \in Y} deg(v)$$

since $deg(v) = x$ for $x \in X$ and $deg(v) = y$ for $y \in Y$, the result follows immediately.

Question 2.18. Which of the following graphs are bipartite: C_n, K_n, E_n, P_n, Q_n?

Answer: Let $n > 1$ since when $n = 1$ all graphs are bipartite. C_n: Since a graph is bipartite iff it contains no odd cycle, C_n is only bipartite when n is even.
K_n: only bipartite when $n = 2$ as when $n \geq 3$, K_n contains a copy of C_3.
E_n: for all n, since we can just take any partition.
Q_n: There are many ways to do this, the simplest way is to notice that we can partition the vertex set into one set containing all vertices with an even number of 1s, and another set containing all vertices with an odd number of 1s. There are no edges within these sets, as two vertices within the each set must differ by at least two coordinates. So Q_n is bipartite for all n.

2.8 Graph Colouring

Question 2.19. For a graph G let

$$\omega(G) = max\{t : G \text{ contains a copy of } K_t\}$$

and

$$\alpha(G) = max\{|A| : A \subseteq V(G) \text{ is an independent set }\}$$

Show that $\chi(G) \geq max\{\omega(G), n/\alpha(G)\}$

Answer: First prove $\chi(G) \geq \omega(G)$. If $K_t \subset G$ where $\omega(G) = t$ then $\chi(G) \geq \chi(K_t) \geq t = \omega(G)$.
Now prove $\chi(G) \geq n/(\alpha(G))$. Note that a colouring seperates $V(G)$ into colour classes, each of which cannot be bigger than $\alpha(G)$. So we have $n = |V(G)| \leq \chi(G) \cdot \alpha(G) \implies \chi(G) \geq n/(\alpha(G))$. Result follows immediately.

Question 2.20. Let G be a graph with chromatic number k. Show that G must have at least $\binom{k}{2}$ edges.

Answer: Chromatic number k means that $V = V_1 \ldots \cup V_2 \ldots \cup \ldots V_k$ is the smallest partition we can get (i.e. we couldn't get $V = V_1 \ldots \cup V_2 \ldots \cup \ldots V_{k-1}$, since then it would be $k-1$-colourable). In order to be k-partite, we need that each partition has an edge going to every other partition, otherwise two partitions have no edges between them and thus can be collapsed into the same partition. Hence this means we have at least $\binom{k}{2}$ edges.

Question 2.21. Let $G = (A, B; E)$ be the bipartite graph with vertex set $A \dot{\cup} B$ where $A = \{a_1, \ldots, a_n\}$ and $B = \{b_1, \ldots, b_n\}$ and edges:

$$E = \{a_i b_j : 1 \leq i, j \leq n, i \neq j\}$$

1. Determine $\chi(G)$ and find an ordering of the vertices so that the greedy algorithm uses exactly $\chi(G)$ colours.

2. Find an ordering of the vertices so that the greedy algorithm uses as many colours as possible.

Answer: Since G is a bipartite graph with size greater than 1, we have that $\chi(G) = 2$. The ordering in which the greedy algorithm gives two colours is $V(G) = \{a_1, a_2, \ldots, a_n, b_1, \ldots, b_n\}$.

For a_1 to a_n, the algorithm gives them all the colour 0, since none are adjacent to eachother. Then from b_1 to b_n the algorithm gives all of these vertices the colour 1, since each vertex is adjacent only to vertices with colour 0.
To get the maximal colouring, we arrange the vertices like $\{a_1, b_1, a_2, b_2, \ldots, a_n, b_n\}$. This gives us n colours: a_1 and b_1 are given the colour 0 since they are non adjacent, then a_2 is given the colour 1 since it is adjacent to b_1. b_2 is also given colour 1 since it is adjacent to a_1 but not a_2. Repeating this process gives colour classes of size two consiting of $\{a_i, b_i\}$ and thus uses n colours. We know that this is maximum since $\Delta(G) + 1 = n - 1 + 1 = n$.

Chapter 3

Matchings

Question 3.1. Let G be a connected bipartite graph with vertex classes X, Y. Show that every edge of G extends to a matching from X to Y iff $|\Gamma(A)| > |A|$ for every $A \subset X, A \neq \emptyset, X$.

Answer: The only if direction follows immediately from Hall's condition. For the if direction, again due to halls condition, we just need to show that if the graph is connected then we dont get $|\Gamma(A)| = |A|$ for every $A \subset X, A \neq \emptyset, X$. Assume there is a matching from X to Y, and there exists a strict non-trivial subset A such that its size is equal to the size of its neighborhood.

Question 3.2. An $n \times n$ Latin square is an $n \times n$ matrix with each entry from $1, \ldots, n$ such that no two entries in the same row or column are the same. Prove that every $r \times n$ latin rectangle may be extended to an $n \times n$ latin square

Answer: sorry

Chapter 4

The Probabilistic Method

4.1 Basics

Question 4.1. Let $G = (V, E)$ be a graph with e edges. Prove that if $k \geq 2$ then there exists a partition of V into k classes: $V = V_1 \dot\cup V_2 \dot\cup \ldots \dot\cup V_k$ such that at least $\lceil (k-1)e/k \rceil$ of the edges in E go between different classes.

Answer: This is very similar to the construction in Prop 4.5. Consider a random partition $V_1 \dot\cup V_2 \dot\cup \ldots \dot\cup V_k$ of V as follows. For each $v \in V(G)$ toss a fair k-sided dice (i.e. outcomes $1 \leq i \leq k$), placing $v \in V_i$ depending on the value of the dice. Evidently all dice throws are independent.
For an edge $uv \in E(G)$ let X_{uv} be such that

$$X_{uv} = \begin{cases} 1, & \text{if } u, v \text{ are in different parts,} \\ 0, & \text{otherwise.} \end{cases}$$

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Question 4.2. Let $e(A)$ denote the number of edges contained in a set of vertices A. Prove that if $G = (V, E)$ is a graph then there is a partition $V = V_1 \cup V_2$ satisfying $e(V_1) + e(V_2) \leq |E|/2$ and $\max_{i=1,2} e(V_i) \leq |E|/3$.

Answer: Done this one just need to write up.

Question 4.3. A set $A \subsetneq \mathbb{N}$ is sum-free if there are no $a, b, c \in A$ satisfying $a + b = c$. Let $p = 3k + 2$ prime satisfying $p > 2maxB$ where $B \subsetneq \mathbb{N}$ and let $T = \{k+1, \ldots, 2k+1\}$

1. Show that T is sum-free modulo p.
2. Let $m \in [p-1]$ be chosen uniformly at random. If $b \in B$ show that $\mathbf{P}[mb \, mod \, p \in T] = \frac{|T|}{p-1} = \frac{k+1}{3k+1} > 1/3$
3. Define $B_m = \{mb \; mod \, p : b \in B\} \subset [p-1]$. Show that $\mathbf{E}[|B_m \cap T|] > |B|/3$
4. Let m^* be chosen such that $|B_{m^*} \cap T|$ is maximised. Explain why its greater than $|B|/3$

5. Let $A = \{b \in B : m^*b \; mod \, p \in T\}$. Explain why A is a sumfree subset of B with more than $|B|/3$ elements.

Answer:

1. Assume that T is not sum free, then $s + t - u = m(3k + 2)$ where $m \in \mathbb{Z}$. Observe that $\max\{s + t - u\} = (2k + 1) + (2k + 1) - (k + 1) = 3k + 1 < 3k + 2$, hence m $= 0$ and we have $s + t = u$. But $\min s + t = 2k + 2 > \max u = 2k + 1$ so we have a contradiction.

2. Note that $p - 1 = 3k + 1$ so $m \in [3k + 1]$. Hence $mb \, mod \, p \in T$ is essentially asking if $m \in T$. $T \subset [3k + 1]$ so since we are sampling uniformly the probability is $\frac{|T|}{p-1}$. $|T| = k + 1$ so $|T|/p - 1 = k + 1/3k + 1 > 1/3$

3. For each b, let X_b be the indicator variable for the event $mb \, mod \, p \in T$. Then we have, by linearity of expectation

$$\begin{aligned}
\mathbf{E}[|B_m \cap T|] &= \mathbf{E}[\sum_{b \in B} X_b] \\
&= \sum_{b \in B} \mathbf{E}[X_b] \\
&> \sum_{b \in B} 1/3 \\
&= |B|/3
\end{aligned}$$

4. Let m^* be the value at which $|B_m \cap T|$ is maximised. By first moment, $\exists \omega_1, \omega_2 \in \Omega$ such that

$$X(\omega_2) > \mathbf{E}[|B_m \cap T|] > |B|/3$$

   Since m^* gives the maximal value,$|B_{m^*} \cap T|$ must be the bigger than any other event. so the inequality holds.

5. A is clearly a subset of B. Assume that $\exists s, t, u \in A$ such that $s + t = u$. Then we have that $m^*s + m^*t = m * u$ mod $p \in T$. The size of p ensure no "wraparound" problems. This implies that T is not sum free mod p which is a contradiction. Hence A is sum-free. Observe also that $|A| = |B_{m^*} \cap T|$.

4.2 Mycelskian Graphs

No questions! woohoo!

4.3 Random Graphs

Question 4.4. What is the expected number of edges in a random graph $H \in \mathcal{G}(n,p)$?

Answer: Let X_{uv} be the indicator variable

$$X_{uv} = \begin{cases} 1, & \text{if } uv \in E(H) \\ 0, & \text{otherwise} \end{cases}$$

Each edge is in $E(H)$ with probability p, so $\mathbb{E}[X_{uv}] = 1 \cdot \mathbb{P}[uv \in E(H)] = p$. There are $\binom{n}{2}$ potential edges; let X be the total number of edges, then by linearity of expectation:

$$\mathbb{E}[X] = \mathbb{E}[\sum_{uv \in E(H)} X_{uv}] = \sum_{uv \in E(H)} \mathbb{E}[X_{uv}] = \binom{n}{2} p$$

Question 4.5. For which $p \in [0,1]$ is it true that $\mathcal{G}(n,p)$ is simply the space of all labeled graphs on vertex set $[n]$ with the uniform probability measure? (i.e. when is generating a graph from $\mathcal{G}(n,p)$ is the same as sampling a graph uniformally at random from the set of all graphs on the vertex set $[n]$)

Answer: Intuitively, the answer is immediately $p = 1/2$ since we want the existence of an edge to be as likely as its non existence, generating all graphs at the same probability. Formally, let X denote the number of edges in a given graph $H \in \mathcal{G}(n,p)$. From before, we know that if we generate a random graph, its expected number of edges is $\mathbb{E}_1[X] = \binom{n}{2}p$. In a uniformally random graph, each of the $\binom{n}{2}$ edges is present in exactly half the graphs. So the expected number of edges in a uniformally sampled graph is $\mathbb{E}_2[X] = \frac{\binom{n}{2}}{2}$ so $p = 1/2$ as the two expectations must be equal.

Question 4.6. What is the expected number of triangles in a random graph $H \in \mathcal{G}(n,p)$

Answer: There are $\binom{n}{3}$ unordered triples of vertices. For some fixed triple $\{i,j,k\}$. Let Y_{ijk} be the indicator variable that all 3 edges forming a triangle are present. Since the presence of an individual edge is independent, defined in the natural way. We have that $\mathbb{E}[X_{ijk}] = p^3$. Let $T = \sum_{ijk} Y_{ijk}$ be the number of triangles. Then by linearity of expectation $\mathbb{E}[T] = \binom{n}{3}p^3$.

4.4 Large Girth and Large Chromatic Number

Question 4.7. If $t \geq 2$, what is the expected number of copies of K_t in a random graph $H \in \mathcal{G}(n,p)$?

Answer: Let X_t be the number of cliques of order t in H. Any given clique has $\binom{t}{2}$ edges so $\mathbb{P}[\text{a given } K_t \text{ is in } H] = p^{\binom{t}{2}}$. Since any t-tuple can give a clique of order t, there are $\binom{n}{t}$ possible ways to choose the vertices for such a clique. Hence by linearity of expectation $\mathbb{E}[X_t] = \binom{n}{t}p^{\binom{t}{2}}$.

Question 4.8. What is the expected number of copies of C_5 in a random graph $H \in \mathcal{G}(n,p)$

Answer: Immediate application of Lemma 4.10 gives $\mathbb{E}[C_5] = \frac{n(n-1)(n-2)(n-3)(n-4)}{10}p^5 = 12\binom{n}{5}p^5$

Chapter 5

Extremal Graph Theory

5.1 Hamilton Cycles

Question 5.1. Consider the proof of Ore's theorem. Why is the process of adding edges to G that do not create a Hamilton cycle guranteed to terminate?

Answer: Eventually the graph will become complete, which surely has a hamilton cycle.

Question 5.2. Deduce Dirac's theorem from Ore's theorem

Answer: If a graph meets Dirac's condition, then it satisfies Ore's condition since if $\delta(v) \geq n/2$ then $d(v) + d(u) \geq n/2 + n/2 = n$. so Dirac's condition guarantees there is a Hamilton cycle.

Question 5.3. Prove that for any $n \geq 3$ the condition $\delta(G) \geq n/2$ is *best possible* (i.e. find a graph G_n with $\delta(G_n) \geq n/2 - 1$ that does not contain a Hamilton cycle).

Answer: To prove for general n, consider the graph $K_{k,n-k}$. The only way there can be a hamilton cycle in a bipartite graph is if both partitions are of equal size, since the path needs to alternate between partitions. So for $n = 2k$ we can take $K_{k+1,k-1}$. $\delta(G) = k - 1 = n/2 - 1$ but has no hamilton cycle. For $n = 2k + 1$ we can take $K_{k,k+1}$, then $\delta(G) = k \geq (n-1)/2 \geq n/2 - 1$

Question 5.4. Prove that any graph of order $n \geq 3$ and size at least $\binom{n-1}{2} + 2$ contains a hamilton cycle. Give an example to prove that this is best possible.

Answer: For the first part, any graph of order n with $\binom{n-1}{2} + 2$ edges must contain a clique of order $n - 1$. All of the nodes in the clique have order $n - 2$. There are $n - 1$ nodes in the clique, so there must be some node not in it, say v. As K_{n-1} has $\binom{n-1}{2}$ edges there must be at least two edges left. As we only allow simple graphs, these edges must be adjacent to v, so $d(v) \geq 2$. Hence Ore's theorem applies and the graph is Hamiltonian. Note that K_{n-1} has $\binom{n-1}{2}$ edges, so if we adjoin a path of length 1 to one of the nodes of K_{n-1} then it has less than $\binom{n-1}{2} + 2$ edges but clearly does not have a hamilton cycle.

Question 5.5. Prove that the hypercube Q_n has a hamilton cycle for each integer

$n \geq 2$.

Answer: Can't use Ore's or Dirac's since $2n \ll 2^n$ as $n \to \infty$. I will sketch a proof via induction. Suppose that we have a hamiltonian cycle for Q_{k-1}. Note that Q_k is just two copies of Q_{k-1} with $k-1$ edges between them. Construct a Hamiltonian cycle as follows. Follow the hamiltonian cycle on the first Q_{k-1} until you are one edge away from reaching the starting vertex. Then use one of the $k-1$ edges to traverse to the other Q_{k-1} and do the same hamiltonian path you have just done but in reverse, again stopping before you return to the starting node for this hamiltonian cycle. As you have done it in reverse, you will be adjacent to the starting vertex of the first hamiltonian cycle (like a mirror!). Traverse back to complete the cycle.

5.2 Forbidden Subgraphs: Mantel's Theorem

Question 5.6. Consider the first proof of Mantel's theorem. Can you adapt this argument to prove the following result: if G is a K_3-free graph of order n and size $e = \lfloor n^2/4 \rfloor - t$, where $t \in \mathbb{N}$, then G contains a bipartite subgraph with at least $e - t$ edges.

Answer: Suppose that G is K_3 free of order n. Let $A \subseteq V$ be the largest indepdent set with $|A| = a$. Note that all edges in G have at least one vertex in $V \setminus A$ otherwise A would no longer be indepedent. Let $E(H)$ be the bipartite subgraph induced by taking the edges which have exactly one vertex in $V \setminus A$. The remaining edges must have both vertices in $V \setminus A$, call these E_{rem}. Then we have $|E(G)| = |E(H)| + |E_{rem}|$. Rearranging and substituting our known values we get:

$$|E_{rem}| = e - |E(H)|$$

Since the graph is K_3-free, we have that each $\Gamma(v)$ is an independent set and thus $d(v) = |\Gamma(v)| \leq a$. Hence we have the following inequality:

$$\sum_{v \in V \setminus A} d(v) = |E(H)| + 2|E_{rem}| \leq n^2/4 = e + t$$

Thus substituting earlier values we have:

$$\begin{aligned} |E(H)| + 2(e - |E(H)|) &\leq e + t \\ \implies -|E(H)| + 2(e) &\leq e + t \\ \implies e - t &\leq |E(H)| \end{aligned}$$

as required

Question 5.7. .

Answer:

1. G must contain an odd length cycle as it is not bipartite.

2. $C_3 \cong K_3$ and it is K_3-free so the smallest odd cycle it can have is of length 5.

3. Let $C = v_1, v_2, \ldots, v_{2k+1}$. If $v \in C$ then obviously it has at most 2 neighbours otherwise we create a K_3. Suppose that there is a $w \in V(G) \setminus C$ with more than two neighbours in C, WLOG call these v_i, v_j, v_l, all of these edges must be non-adjacent, as otherwise we create a K_3. There is no way to pick 3 nodes from C_5 such that they are all non adjacent (draw a picture to convince yourself). Hence in this scenario we must have at least $|C| \geq 7$. If $|C| = k$ then we can form a cycle of length $k-2$ by using the extra edges given by the neighbours. Not going to prove this explicitly.

4. Let e be the number of edges from $G \setminus C$ to C. Let $|G \setminus C| = M$. Note that $C \subseteq G$ so $M \geq 0$. Using our two bounds we get:

$$e > \sum_{v \in G \setminus C} \delta(v) = 2n/5 \cdot M$$

$$e \leq 2M$$

5. Putting those two together we get $M2n/5 < e \leq 2$ which implies that $n < e \leq 5$. This is obviously impossible when $n \geq 5$, but we must have $n \geq 5$ as parts 1 and 2 guarantee the existence of a cycle of at least length 5.

Question 5.8. .

Answer: C_5 suffices.

Question 5.9. .

Answer: Let G be any graph of order n split into two classes A and B with sizes as equal as possible. Either way we have $|A| = |B| = n/2$ or WLOG $|A| = (n+1)/2$ and $|B| = (n-1)/2$. Either way, the maximum number of edges between them is $\lfloor n^2/4 \rfloor$. The total number of possible edges in G is $\binom{n}{2}$ so given any edge e, the probability that e is one of the crossing edges is

$$P(e \text{ crosses}) = \frac{\lfloor n^2/4 \rfloor}{\binom{n}{2}} = \frac{2\lfloor n^2/4 \rfloor}{n(n-1)}$$

If we let X be a random variable measuring the amount of edges that cross, by linearity of expectation we have:

$$E[X] = \frac{2m\lfloor n^2/4 \rfloor}{n(n-1)}$$

Hence by the first moment there must be a bipartite subgraph with at least this many crossing edges.

5.3 Forbidden Subgraphs: Turán's Theorem

Question 5.10. How many edges does $T_4(14)$ contain? Does $T_4(14)$ contain a subgraph isomorphic to $T_3(10)$?

Answer: $T_4(14)$ is a complete 4-partite graph wherein two classes have 3 vertices, and the other 2 have 4 vertices. Hence there are 73 edges. $T_3(10)$ is just $T_4(14)$ but with only one vertex class containing four vertices. So yes.

Question 5.11. Give a formula for $t_r(n)$ in terms of r, n and $t_r(n-1)$

Answer: Let $T_r(n)$ be the r-partite graph with b vertex classes of order $\lfloor n/r \rfloor$ and $r-b$ classes of order $\lceil n/r \rceil$ where b satisfies $n = b\lfloor n/r \rfloor + (r-b)\lceil n/r \rceil$. Within $T_r(n)$ we can find a copy of $T_r(n-1)$ by deleting one of the vertices. If we specifically delete a vertex from one of the $\lceil n/r \rceil$ sized classes, we don't get any reshuffling (convince yourself of this!). So by deleting one vertex we lose the $b\lfloor n/r \rfloor$ edges from the smaller classes and $(r-1-b)\lceil n/r \rceil$ from the larger classes. Hence:

$$\begin{aligned} t_r(n) &= t_r(n-1) + b\lfloor n/r \rfloor + (r-1-b)\lceil n/r \rceil \\ &= t_r(n-1) + n - \lceil n/r \rceil \end{aligned}$$

Question 5.12. Let G be a graph on $n \geq 5$ vertices and with minimal degree $\delta(G) > \frac{3n}{4}$. Show that G contains a copy of K_5

Answer: The trick with this question is to show that $|E(G)| \geq ex(n, K_5) = t_4(n)$. How many edges are in $T_4(n)$? The number of edges is maximised when n is divisible by 4, so assume this is the case. Then $t_4(n) = 3 \cdot (n/4)(n/4) + 2 \cdot (n/4)(n/4) + (n/4)(n/4) = 6n^2/16 = 3n^2/8$. The total number of edges in G is at least $3n^2/8$ due to the minimal degree and handshake lemma. Hence the number of edges in G is at least the turan number so must contain a K_5.

Question 5.13. .

Answer: Suppose that G is not bipartite, and let $C = v_1, \ldots, v_{2k+1}$ be the shortest odd cycle. Note that this must exist and $|C| = 2k+1 \geq 5$ by question 5.7. . Obviously the number of edges in C is $2k+1$. In $G \setminus C$. Since $G \setminus C$ is still K_3 free we can use mantel's to bound the number of edges in $G \setminus C$ to be $\lfloor (n - |C|)^2/4 \rfloor$. Then from 5.7 each of the vertices in $G \setminus C$ has at most 2 neighbours in C. So the max number of vertices between $G \setminus C$ and C is $2(n - |C|)$. Putting all this together we get:

$$|E(G)| \leq |C| + 2(n - |C|) + \lfloor (n - |C|)^2/4 \rfloor$$

Substituting $|C| \geq 5$, rearranging, and calculating that $t_2(n-1) = \lfloor (n-1)^2/4 \rfloor$ we get that $|E(G)| \leq t_2(n-1) + 1$ (trust me! I just didn't want to write all the algebra). Thus we have a contradiction.

5.4 Digression: Double counting

no questions!!!!

5.5 Asymptotics: Turán density

Question 5.14. Prove that $\pi(K_{1,t}) = 0$ for any $t \geq 1$.

Answer: The existence of $K_{1,t}$ as a subgraph is equivalent to finding a vertex of degree at least t, so being $K_{1,t}$ free is equivalent to having max degree $t-1$. The handshake lemma then implies that $ex(n, K_{1,t}) \leq (t-1)n/2$. Hence

$$\pi(K_{1,t}) = \lim_{n\to\infty} \frac{ex(n, K_{1,t})}{\binom{n}{2}} = \lim_{n\to\infty} \frac{(t-1)n/2}{(n-1)n/2} = 0$$

Question 5.15. Prove that $\pi(C_{2r}) = 0$ for any $r \geq 1$

Answer: Not sure how to do this without stealing from later in the course. Erdos–Stone says that since $\chi(C_{2r}) = 2$ we have that

$$\pi(C_{2r}) = 1 - \frac{1}{2-1} = 0$$

5.6 Bipartite forbidden subgraphs

5.7 Erdos–Stone: the fundamental theorem of extremal graph theory

Chapter 6

Families of sets: Chains, antichains and intersection problems

6.1 Posets,chains and antichains

6.2 LYM-inequality

6.3 Intersecting Families

Question 6.1. Let $n \geq 4$, give three examples of non-isomorphic intersecting families of sets in $\mathcal{P}([n])$ of size 2^{n-1}. Note: two families of sets $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}([n])$ are isomorphic iff there is a bijection $f : [n] \to [n]$ such that $\mathcal{B} = \{f(A) : A \in \mathcal{A}\}$.

Answer:

6.4 The linear algebra method

No questions!

6.5 L-intersecting families

Chapter 7

Ramsey Theory

7.1 Ramsey's theorem

Question 7.1. Show that $R(3,5) \leq 14$

Answer: Suppose that we have a random red-blue colouring of K_{14}. Suppose that we have some vertex v with $d_{red} \geq 5$. Either one of the edges in $\Gamma_{red}(v)$ are red, and thus we automatically get a red K_3 with v or they're all blue, and we get a blue K_5. Now suppose that for all v we have $d_{red}(v) \leq 4$, then by symmetry we have that $d_{blue}v \geq 9$. Since $R(3,4) = 9$, any way that we red-blue colour $\Gamma_{blue}(v)$ we get a red K_3 or a blue K_4. If we get a red K_3 then we are done, if we get a blue K_4 then we can form a blue K_5 with v.

Question 7.2. Show that $R(5,5) \leq 64$

Answer: From the proof of Ramsey's theorem, we have that $R(s,t) \leq R(s-1,t) + R(s,t-1)$. So $R(5,5) \leq R(4,5) + R(5,4) = 2R(4,5)$. Each $R(4,5) \leq R(3,5) + R(4,4) \leq 14 + 18 = 32$. Hence $R(5,5) \leq 2 \cdot 32 = 64$.

Question 7.3. Show that $R(5,5) \leq 63$

Answer: Sorry. I think we can actually get the bound that $R(s,t) \leq R(s-1,t) + R(s,t-1) - 1$ but I dont know how to prove it. From there it follows immediately.

Question 7.4. .

Answer: Assume the graph containing all the red edges is disconnected. This means we can split the entire graph into (at least) two components without any red ages between them. Let these components be called A and B, and pick arbitrary $a \in A$, $b \in B$. There must be an edge between a and b since we are in a complete graph, and this edge must be blue. This is true for all $a \in A$ and $b \in B$. So the blue subgraph is connected.

7.2 Ramsey numbers: lower bounds and more colours

7.3 Ramsey theory in the integers

7.4 VDW theorem